

The Dynamic Circulation

Navier–Stokes · the elastic wall and the pulse wave · the two-element
Windkessel

From Physics to Physiology · Lecture 2
MSc course material · 2026

Scope of this lecture

Lecture 1

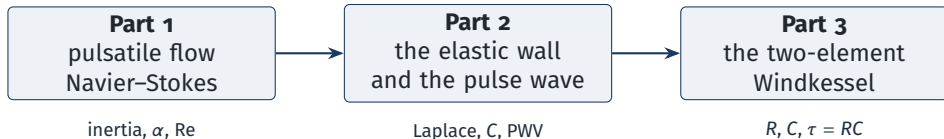
steady flow in rigid tubes

- nothing depends on time $\rightarrow$ no acceleration of blood
- vessel geometry fixed $\rightarrow$ no storage of volume

Lecture 2

two idealizations removed

- **Time**. The heart beats: blood accelerates and decelerates $\rightarrow$ inertia $\rightarrow$ Navier–Stokes
- **Elasticity**. The wall stretches: it stores volume and carries the pulse as a wave



What Lecture 1 established

- Conservation of volume: Kirchhoff at junctions; $Q = vA$
- Poiseuille: resistance concentrated in arterioles (fourth power of the radius)
- Series and parallel networks; pressure dividers
- Wall shear stress: the quantity sensed by the endothelium

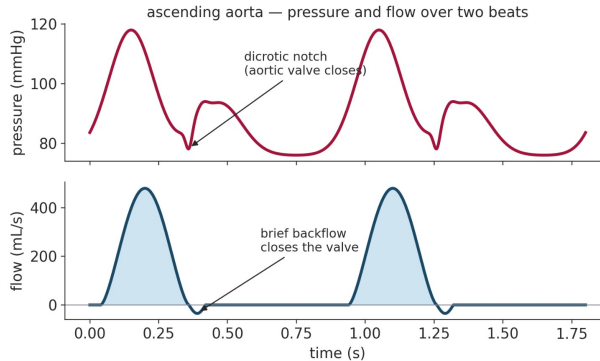
$$R = \frac{8\mu L}{\pi r^4}, \quad \Delta P = R Q$$

$$\tau_w = \frac{4\mu Q}{\pi r^3}$$

Key idea

A complete toolkit — for a world without time and without elastic walls. Both are now admitted.

The empirical starting point



Simultaneous pressure and flow in the ascending aorta over two cardiac cycles. Every feature of these two curves acquires a name and an equation today.

Three features that require explanation

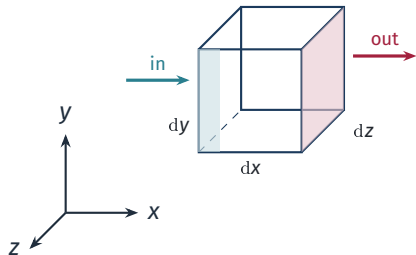
- **The flow curve** — large amplitude, short duration: ejection peaking at 400–500 mL/s, a brief backflow transient, then zero flow for two thirds of the cycle.
- **The pressure curve** — never below ~75 mmHg despite zero inflow; the decay is smooth and closely exponential → something *stores* volume in systole and *releases* it in diastole (Part 3).
- **The incisura** — the notch marking aortic valve closure.

systolic ≈ 120 mmHg · diastolic ≈ 80 mmHg · $PP = SBP - DBP \approx 40$ mmHg · $MAP \approx DBP + PP/3$

Part 1 · Pulsatile flow and the Navier–Stokes equations

Control-volume balances → the full system → the classical laws as limiting cases → non-dimensionalization → the Womersley and Reynolds numbers

Conservation of mass: a fixed control volume



Eulerian viewpoint: the box is fixed in space;
the fluid streams through it.

The two flux terms *almost* cancel: what survives is the derivative. The y- and z-face pairs contribute analogously.

Only two mechanisms change the mass inside:

- 1 **Accumulation** (volume fixed; only density may change)
- 2 **Net flux** through each pair of opposite faces

$$\frac{\partial(\delta m)}{\partial t} = \delta V \frac{\partial \rho}{\partial t}$$
$$\underbrace{\rho u \, dy \, dz}_{\text{in at } x} - \underbrace{\left[\rho u + \frac{\partial(\rho u)}{\partial x} dx \right] dy \, dz}_{\text{out at } x+dx} = -\frac{\partial(\rho u)}{\partial x} \delta V$$

The mass balance, assembled

Balance. Accumulation inside = minus the net outflow through the six faces:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad \text{continuity equation — valid for any fluid}$$

Blood is incompressible to an excellent approximation ($\rho = 1060 \text{ kg/m}^3$, constant):

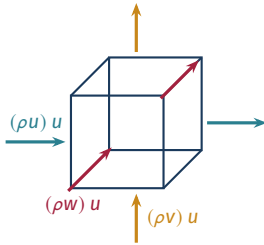
$$\nabla \cdot \mathbf{v} = 0 \quad \Longleftrightarrow \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Key idea

Kirchhoff's junction law (Lecture 1) and $\nabla \cdot \mathbf{v} = 0$ are the *same statement at different magnifications*: what flows in must flow out — whether “in” is a branch of the vascular tree or a face of an infinitesimal cube.

Conservation of momentum I: accumulation and transport

Repeat the bookkeeping for the x-momentum per unit volume, ρu , before any force acts.



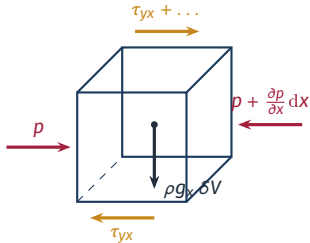
- **Accumulation**, exactly as for mass
- **Convective transport**. Fluid crossing a y-face at velocity v carries its x-momentum ρu with it; only differences survive

$$\frac{\partial(\rho u)}{\partial t} \delta V$$
$$\left[\frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho v u)}{\partial y} + \frac{\partial(\rho w u)}{\partial z} \right] \delta V$$

x-momentum is carried through *all* six faces.

These become the **convective acceleration** $(\mathbf{v} \cdot \nabla)\mathbf{v}$ — present even in steady flow — and the **local acceleration** $\partial\mathbf{v}/\partial t$ — the heartbeat term, identically zero in Lecture 1.

Conservation of momentum II: the forces



Pressure: normal, compressive. Viscous stress: drag of the neighbouring fluid on each face.

- **Body force** on the volume: $\rho f_x \delta V$ (gravity, $f = g$)
- **Pressure** on the x-faces — only the gradient survives:

$$p \, dydz - \left[p + \frac{\partial p}{\partial x} dx \right] dydz = -\frac{\partial p}{\partial x} \delta V$$

- **Viscous stresses** — normal τ_{xx} on the x-faces, shear τ_{yx} , τ_{zx} on the others:

$$\left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \right] \delta V$$

The x-momentum equation

Assemble **accumulation** + **flux** = **forces**; expand the products, subtract $u \times (\text{continuity})$, and insert the Newtonian law $\tau = \mu \times (\text{rate of strain})$:

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x$$

Key idea

Every surface quantity entered the balance as a *difference between opposite faces*, so every force appears as a *derivative*. That is why the law of motion of a fluid is a partial differential equation.

Assumptions carried from here on: continuum, incompressible, Newtonian, laminar — the same simple fluid as in Lecture 1.

The full system in Cartesian coordinates

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x$$

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho g_y$$

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g_z$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Four scalar equations · four unknown scalar fields u , v , w , p

The compact form — and the name of every term

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \rho(\mathbf{v} \cdot \nabla)\mathbf{v} = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}, \quad \nabla \cdot \mathbf{v} = 0$$

$$\rho \frac{\partial \mathbf{v}}{\partial t}$$

unsteady inertia

the heart-beat term
new today

$$\rho(\mathbf{v} \cdot \nabla)\mathbf{v}$$

convective inertia

the stenosis term
nonlinear

$$-\nabla p$$

pressure gradient

only differences push

$$\mu \nabla^2 \mathbf{v}$$

viscous friction

diffusion of momentum

$$\rho \mathbf{g}$$

gravity

body force

Left-hand side: $\rho \times$ (acceleration of a fluid particle) — Newton's $F = ma$ per unit volume, evaluated at a fixed point in space.

The two accelerations must be kept distinct



local: same point,
velocity changes in time

$$\frac{\partial \mathbf{v}}{\partial t}$$

the flow field itself changes: **the heart beats**



convective: particle moves
to where velocity differs

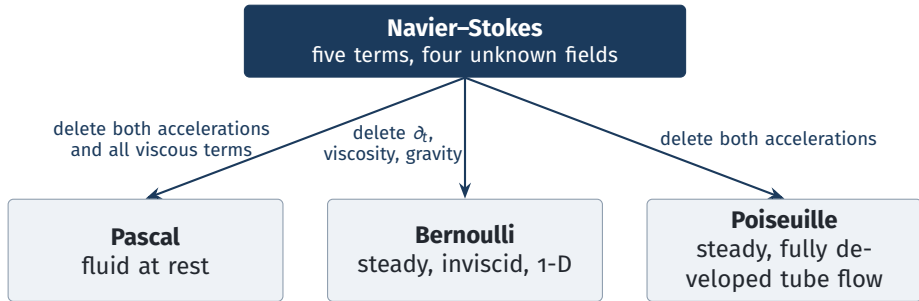
$$(\mathbf{v} \cdot \nabla) \mathbf{v}$$

nonzero in a perfectly steady nozzle: **the stenosis**

Key idea

The convective term makes the system nonlinear: no general analytic solution. A straight tube is one of the few solvable geometries — hence Poiseuille's law. Everything else: CFD.

The classical laws are limiting cases

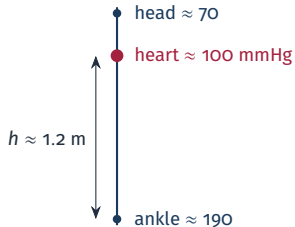


The discipline required: state the *physical justification* for every deletion.

Limiting case 1 · Fluid at rest → Pascal

Delete: $\mathbf{v} \equiv 0$ everywhere and always → both accelerations and the entire viscous term vanish.

$$\nabla p = \rho \mathbf{g} \quad \implies \quad p = p_0 + \rho g h \quad (h: \text{depth below the reference level})$$



The standing subject. Ankle arteries sit ~1.2 m below the heart:

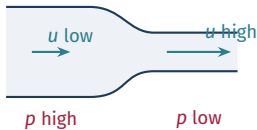
$$\rho g h \approx 1060 \times 9.81 \times 1.2 \approx 12.5 \text{ kPa} \approx 94 \text{ mmHg}$$

— with no flow required.

Limiting case 2 · Steady, inviscid, one-dimensional → Bernoulli

Delete: the time derivative (steady), the viscous term (inviscid), gravity. **Retain:** convective inertia against the pressure gradient.

$$\rho u \frac{du}{dx} = -\frac{dp}{dx} \quad \Rightarrow \quad dp = -\rho u du \quad \Rightarrow \quad p + \frac{1}{2}\rho u^2 = \text{const}$$



Pressure falls where the fluid accelerates: a *lossless* exchange between pressure and kinetic energy — friction was deleted by assumption.

In the clinic: jets and valves. The simplified rule $\Delta P = 4v^2$ (mmHg, m/s) quantifies an aortic-stenosis jet from Doppler velocity alone.

Limiting case 3 · Steady, fully developed tube flow → Poiseuille

Delete: both accelerations (steady, and fully developed so nothing changes along the axis). **Retain:** pressure gradient against viscous friction.

$$0 = -\frac{dp}{dx} + \mu \frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right) \quad \Rightarrow \quad Q = \frac{\Delta P}{R}, \quad R = \frac{8\mu L}{\pi r^4}$$



A cylinder is one of the very few geometries in which the remaining balance can be integrated by hand. The result is the whole of Lecture 1.

In the clinic: vascular resistance, SVR, arteriolar control.

Poiseuille and Bernoulli side by side

	Poiseuille	Bernoulli
Flow type	laminar, steady, developed; no-slip wall	inviscid, steady, along a streamline
Terms dropped	both accelerations (all inertia)	viscosity (and the unsteady term)
Force balance	friction vs. pressure gradient	convective inertia vs. pressure gradient
Geometry	straight, rigid, uniform tube	converging / diverging passages
Law	$Q = \Delta P/R$, $R = 8\mu L/\pi r^4$	$p + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$
In the clinic	vascular resistance, SVR	valve gradients $\Delta P = 4v^2$; the bQ^2 term of FFR

Neither law is more correct: each is Navier–Stokes in its own regime. A stenosis needs *both* ($\Delta P = aQ + bQ^2$). In the aorta *neither* applies cleanly: **unsteady inertia dominates** — the next topic.

Non-dimensionalization: let the equation report its own regime

Rescale every variable by a characteristic quantity of the problem — for an artery:

$$x = Lx^*, \quad \mathbf{v} = V\mathbf{v}^*, \quad t = t^*/\omega, \quad p = \rho V^2 p^*$$

L : vessel radius · V : mean speed · $\omega = 2\pi f$: heart pulsation · ρV^2 : inertial pressure scale

Substitute into Navier–Stokes and divide by the viscous scale $\mu V/L^2$. Every starred quantity is $O(1)$ by construction; all the physics is now in the coefficients:

$$\underbrace{\frac{\rho\omega L^2}{\mu}}_{\alpha^2} \frac{\partial \mathbf{v}^*}{\partial t^*} + \underbrace{\frac{\rho V L}{\mu}}_{\text{Re}} (\mathbf{v}^* \cdot \nabla^*) \mathbf{v}^* = -\text{Re} \nabla^* p^* + \nabla^{*2} \mathbf{v}^*$$

Two numbers govern the whole circulation

Womersley number

$$\alpha^2 = \text{Wo}^2 = \frac{\rho \omega L^2}{\mu}$$

unsteady inertia / friction

Reynolds number

$$\text{Re} = \frac{\rho VL}{\mu}$$

convective inertia / friction

- $\alpha \ll 1$ and $\text{Re} \ll 1$: friction dominates — Poiseuille survives
- $\alpha \gg 1$: the unsteady term dominates — Lecture 1 no longer applies
- Nothing has been solved; the equation now identifies its own regime.

Wo (dimensional-analysis literature) and α (Womersley's notation) denote the same quantity; α is used from here on.

The Womersley number, vessel by vessel

For a cylindrical vessel the characteristic length is the radius, $L = r$:

$$\alpha = r \sqrt{\frac{\omega \rho}{\mu}}$$

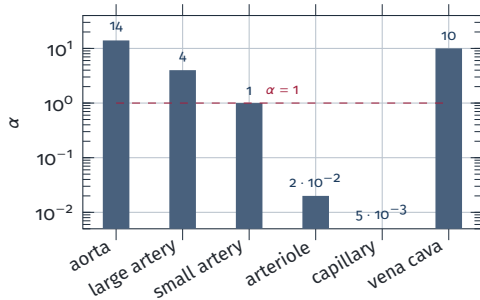
Aorta ($r = 10 \text{ mm}$, $f = 1 \text{ Hz}$)

$$\alpha = 0.010 \sqrt{6.3 \times 3 \times 10^5} \approx 14$$

Arteriole ($r = 15 \mu\text{m}$)

$$\alpha \approx 0.02$$

Three orders of magnitude apart, because $\alpha \propto r$.



Three regimes, one criterion

$$\alpha \ll 1$$

friction dominates

Poiseuille parabola at every instant, tracking the instantaneous gradient. **Lecture 1 remains valid.**

$$\alpha \approx 1-3$$

transitional

The profile lags the gradient and begins to flatten.

$$\alpha \gg 1$$

unsteady inertia dominates

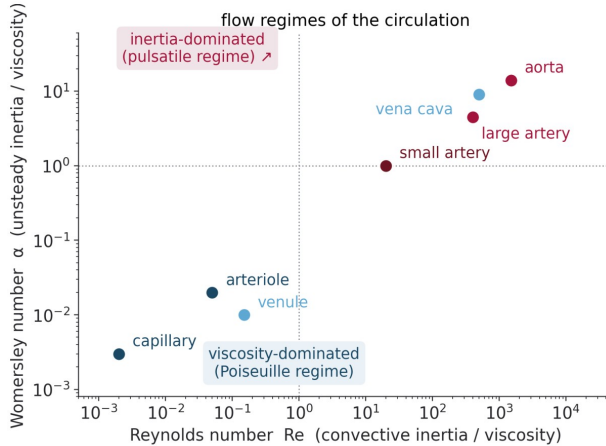
Blunt plug core, thin frictional layer at the wall, *flow out of phase* with the gradient.

Key idea

A single number determines where the formulas of Lecture 1 remain applicable: the microcirculation never leaves that regime; the aorta never enters it.

α is not a Reynolds number: α weighs unsteady inertia, Re weighs convective inertia — both against friction.

The circulation on the $Re-\alpha$ plane

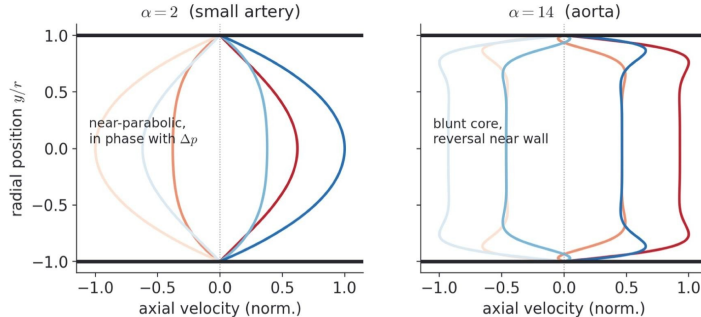


Upper right — aorta. Both inertial terms significant: waves, plug flow, transient turbulence. The analytically difficult regime.

Lower left — microcirculation. Poiseuille is exact for practical purposes.

Every element of blood traverses this map on every circuit: seven orders of magnitude on both axes.

Womersley's oscillatory velocity profiles (1955)



- small α : a parabola at every instant, scaled by the instantaneous gradient
- large α : the core moves as a plug; friction confined to a thin wall layer
- near-wall fluid responds fastest — it may already *reverse* while the core still moves forward

Two consequences for the aorta ($\alpha \approx 14$)

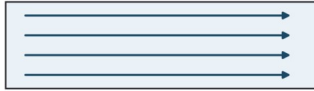
- 1 Wall shear stress is not $4\mu Q/\pi r^3$.** That formula assumes a parabolic profile; in the aorta the profile is blunt, the wall layer thin, and the instantaneous shear is larger and oscillates in sign.
- 2 Measured profiles are blunt, not parabolic.** Ultrasound and phase-contrast MRI in the aorta show plug-like profiles — a direct visual confirmation of large α .

Key idea

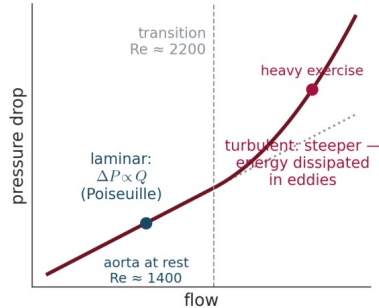
Whenever an equation is used in a large artery, the assumption “parabolic profile” must be checked against α first.

Laminar and turbulent flow: the Reynolds criterion

laminar — ordered layers, silent ($Re < 2200$)



turbulent — erratic eddies, audible ($Re > 2200$)



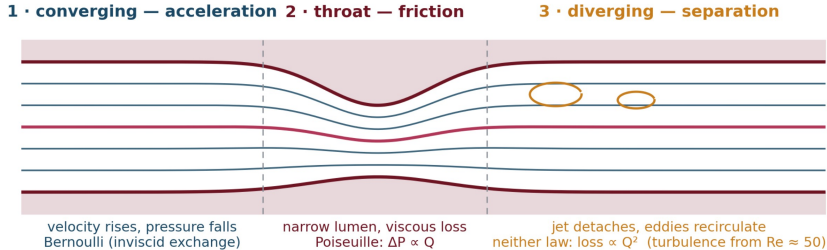
$$Re_D = \frac{\rho v D}{\mu}, \quad Re_c \approx 2200$$

Diameter convention for the transition problem — do not mix with the radius-based definition.

Aorta at rest: $CO = 6 \text{ L/min}$, $r = 1.5 \text{ cm} \rightarrow v \approx 15 \text{ cm/s}$, $Re \approx 1400$ — laminar in the mean; a healthy artery is silent.

Peak-systolic Re transiently exceeds 2200, but one beat is too short for turbulence to develop; *decelerating* flow in late systole is the vulnerable phase.

Worked example · an arterial stenosis in three zones



1 · Converging entrance
Bernoulli: velocity rises,
pressure falls. No loss yet.

2 · Throat
Poiseuille: narrow lumen, high
shear, viscous loss $\propto Q$, $\propto 1/r^4$.

3 · Diverging exit
Separation: the jet detaches,
eddies dissipate the kinetic
energy. Loss $\propto Q^2$.

$$\Delta P = a Q + b Q^2 \quad \text{viscous term} + \text{separation (kinetic) term}$$

Worked example · the arithmetic of a stenosis

Coronary lesion: $a = 2.0 \text{ mmHg}/(\text{mL/s})$, $b = 1.2 \text{ mmHg}/(\text{mL/s})^2$, $\bar{P}_a = 95 \text{ mmHg}$

At rest, $Q \approx 1 \text{ mL/s}$:

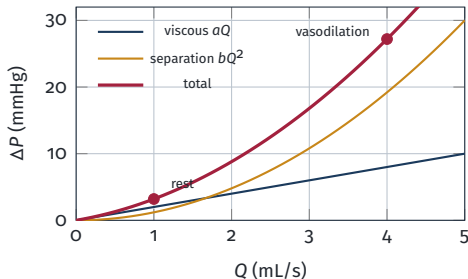
$$\Delta P = 2.0(1) + 1.2(1)^2 = 3.2 \text{ mmHg}$$

$\bar{P}_d \approx 91.8 \text{ mmHg}$ — haemodynamically silent.

At maximal vasodilation, $Q \approx 4 \text{ mL/s}$:

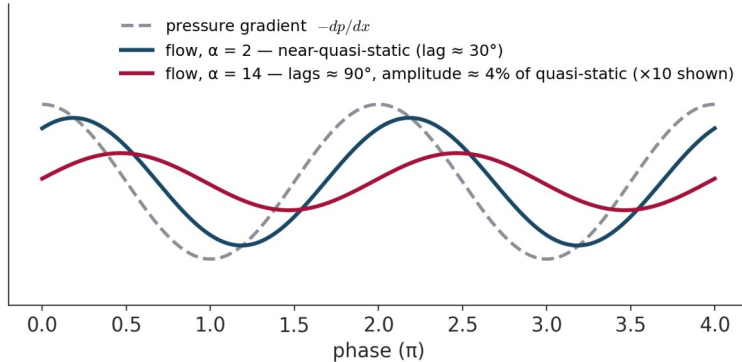
$$\Delta P = 2.0(4) + 1.2(4)^2 = 8.0 + 19.2 = 27.2 \text{ mmHg}$$

$\bar{P}_d \approx 67.8 \text{ mmHg}$, $\text{FFR} = \bar{P}_d / \bar{P}_a \approx 0.71 < 0.80$.



Quadrupling Q multiplies the viscous term by 4 and the separation term by 16 (share 38% → 71%). This asymmetry — not anatomy alone — is why FFR is measured at maximal vasodilation.

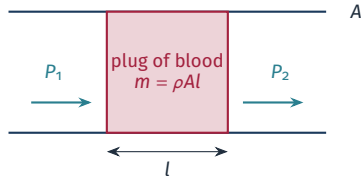
Phase lag: flow runs late



$\alpha = 2$ — nearly quasi-static: flow follows the gradient with a lag of $\approx 30^\circ$.

$\alpha = 14$ — inertia delays the response: flow peaks $\approx 90^\circ$ after the gradient, amplitude attenuated to $\sim 4\%$ of the Poiseuille value.

Inertance: giving the lag a lumped name



$$\Delta P = L \frac{dQ}{dt}, \quad L = \frac{\rho l}{A}$$

inertance

resistance R opposes *flow*

inertance L opposes *change of flow*

Electrical analogue: the inductor.

Newton for the plug: $(P_1 - P_2)A = \rho Al \, dv/dt$

Returns in Part 3 as the physical origin of the characteristic impedance Z_c of the proximal aorta.

Checkpoint · Part 1

- 1 Which term of Navier–Stokes is zero everywhere in Lecture 1, and why?
- 2 Derive Pascal's law from the Navier–Stokes system, naming every deleted term and its justification.
- 3 Compute α for a coronary artery ($r = 1.5$ mm) at 70 bpm. Is Poiseuille's law applicable?
- 4 A murmur is heard over the femoral artery of a resting patient. Why is this never normal?
- 5 In the aorta, does flow lead or lag the pressure gradient, and by approximately how much?

Break

10 minutes

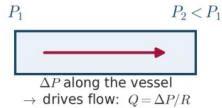
Next: the elastic wall and the pulse wave

Part 2 · The elastic wall and the pulse wave

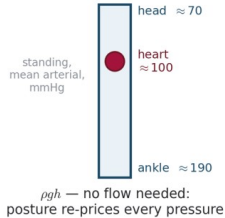
Three pressure differences → wall composition → stress, strain, modulus
→ Laplace → equilibrium and its loss → compliance → wave speed →
reflection and ageing

Three pressure differences, three distinct roles

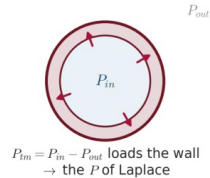
driving pressure



hydrostatic pressure



transmural pressure

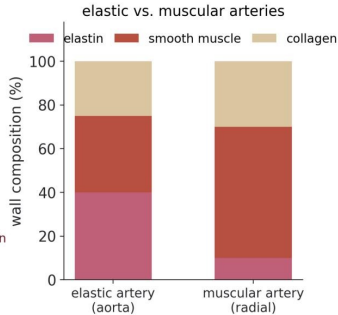
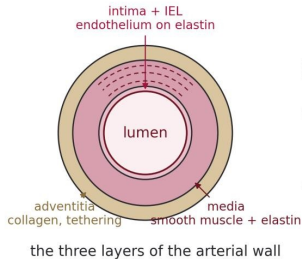


Driving — axial ΔP along the vessel. Drives flow: $Q = \Delta P / R$. A fraction of a mmHg over 10 cm of healthy aorta.

Hydrostatic — ρgh along gravity. Needs no flow (Pascal). Posture rescales every pressure: heart 100, ankle 190, head 70 mmHg.

Transmural — inside minus outside, across the wall. Distends the vessel. **This is the P of Laplace's law and of every compliance measurement.**

Composition of the arterial wall



Intima — endothelium on an elastic sheet: the shear sensor of Lecture 1.

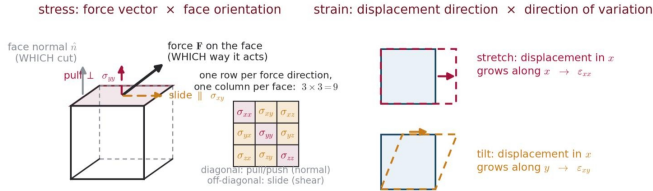
Media — smooth muscle and elastin lamellae: the mechanics of this lecture.

Adventitia — collagen, tethering the vessel.

Elastic arteries (aorta, carotids) store volume: the Windkessel lives here.

Muscular arteries, arterioles regulate calibre: the resistance of Lecture 1 lives here.

Stress and strain require two directions



- **Stress:** (force direction) $\times$ (orientation of the cut face) $\rightarrow$ nine components σ_{ij} ; normal component perpendicular to the face, shear component parallel to it.
- **Pressure** is the isotropic special case: force always perpendicular, equal in every direction — nine components collapse to one scalar. That is why Lecture 1 needed no indices.
- In a vessel all quantities align with three natural directions: **circumferential** (hoop) σ_θ , **axial** σ_a , **radial** σ_r . No indices needed again.

Stress, strain, elastic modulus — the working definitions



$$\varepsilon = \frac{L - L_0}{L_0}$$

strain: relative extension, dimensionless

$$\sigma = \frac{F}{A}$$

stress: force per cross-section, N/m^2

$$\sigma = E \varepsilon$$

Hooke's law; E the elastic modulus

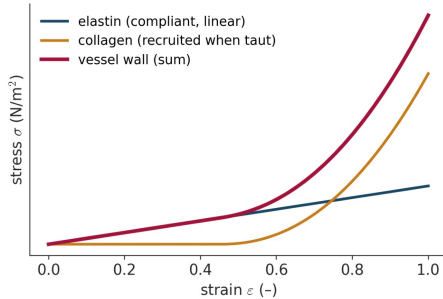


Orders of magnitude: steel $E \approx 200 \text{ GPa}$; aortic wall $E \approx 0.5 \text{ MPa}$ — 4×10^5 times lower.

Key idea

Stress and pressure share units (Pa) but are different objects: pressure is a scalar in the fluid; stress is directional and lives in the wall. The bridge between them is Laplace's law.

Elastin and collagen: a two-fibre composite



Elastin — compliant ($E \approx 0.5$ MPa), linear over large strains, loaded from the first micrometre: the low-pressure spring.

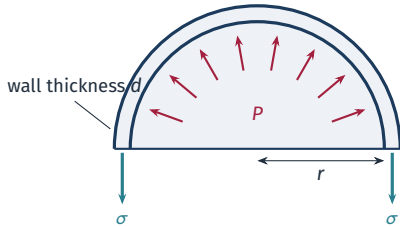
Collagen — very stiff ($E \approx 1$ GPa, $\sim 2000\times$ elastin) but crimped at rest; recruited once straightened, at ~ 10 – 20% strain.

Progressive recruitment $\rightarrow$ the **J-shaped stress-strain curve**: compliant at low strain, stiffening steadily.

Key idea

The modulus of the wall is a property of the *working point*, not a material constant. Ageing: elastin fragments, collagen cross-links — the J-curve steepens at lower strain.

Laplace's law · the equatorial cut



Thin-walled sphere, $d \ll r$, transmural pressure P .
What holds the upper half-dome in equilibrium?

Distending force. Pressure across the cut plane acts on the *projected disc*, not the curved dome:

$$F_p = P \pi r^2$$

Restraining force. Wall stress across the cut ring of wall material:

$$F_w = \sigma \cdot 2\pi r d$$

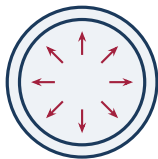
Laplace's law · the result

$$F_p = F_w \quad \implies \quad \sigma = \frac{Pr}{2d} \quad (\text{sphere})$$

- At a given pressure, a larger radius or a thinner wall means higher wall stress.
- Wall *tension* $T = Pr$ is $\sim 3000\times$ larger in the aorta than in a capillary, yet the *stress* is comparable: thickness scales with radius — wall composition is conserved along the tree. A single-cell wall suffices for a capillary.
- Applies to every hollow organ: vessels, heart chambers, alveoli, bladder, tympanic membrane, uterus.

The cylinder: two sections, two stresses

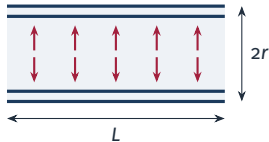
Transverse section



pressure on disc πr^2 , wall ring $2\pi r d$

$$P\pi r^2 = \sigma_a 2\pi r d \Rightarrow \sigma_a = \frac{Pr}{2d}$$

Longitudinal section



pressure on rectangle $2rL$, two strips $2dL$

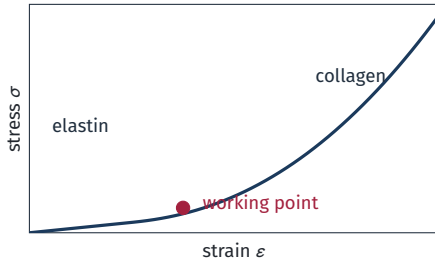
$$P 2rL = \sigma_\theta 2dL \Rightarrow \sigma_\theta = \frac{Pr}{d}$$

$\sigma_\theta = 2 \sigma_a$: the **hoop stress is twice the axial stress** — from geometry alone, independent of the material.

Orders of magnitude — and why arteries sit on the J-curve

The aorta at 100 mmHg

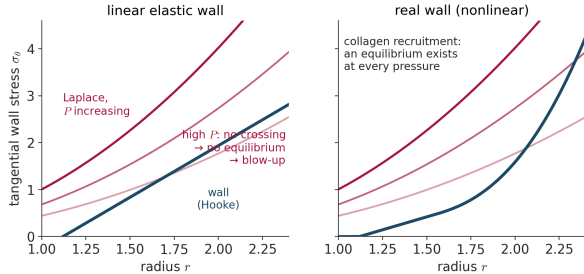
$P \approx 13 \text{ kPa}$, $r/d \approx 6 \rightarrow \sigma_\theta = Pr/d \approx 80 \text{ kPa}$, on a wall of $E \approx 0.5 \text{ MPa} \rightarrow \text{strain } \varepsilon \approx 15\%$.



Arteries operate at *large* strain, on the bend of the J-curve where the load transfers from elastin to collagen.

This is not accidental: it is the **design point**. Below the bend the vessel is a cheap volume store; above it, collagen prevents rupture.

Equilibrium: Laplace load versus elastic response



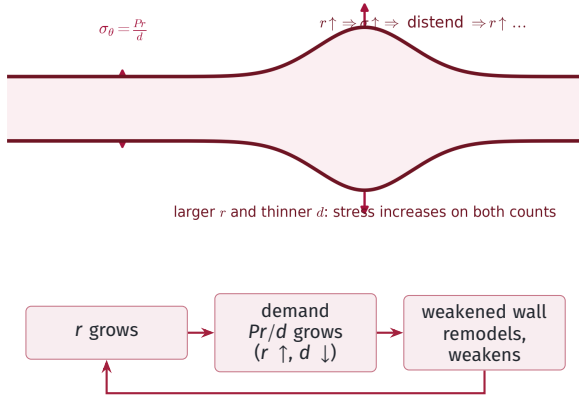
Two curves against radius r :

- **demand:** $\sigma = Pr/d(r)$; the wall thins as it stretches ($d \propto 1/r$), so demand grows as Pr^2
- **supply:** the material's stress-strain curve, via $\varepsilon = (r - r_0)/r_0$

Intersection = equilibrium radius. Stability: the supply must rise *faster* than the demand beyond the crossing.

Key idea. Linear material: above some pressure the Pr^2 demand exceeds the supply everywhere — no intersection, unlimited inflation. **The J-curve is the stability requirement of every pressurised vessel.**

Loss of equilibrium: the aneurysm



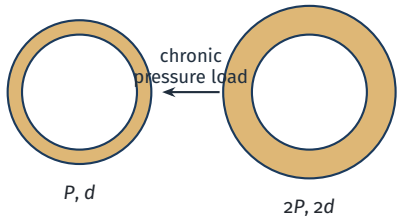
In a degenerated wall — elastin fragmented, collagen disorganised — the supply curve flattens.

Laplace's law becomes the **mechanical driver of the disease**: growth is self-accelerating.

An abdominal aortic aneurysm at 5.5 cm has roughly tripled its wall stress; 5.5 cm is the conventional surgical threshold.

Same physics in another organ: the dilated failing ventricle (Lecture 3).

Laplace read constructively: compensated hypertrophy



$$\sigma = \frac{Pr}{d} : P \uparrow, d \uparrow \Rightarrow \sigma \text{ unchanged}$$

**aneurysm
hypertrophy**

$r \uparrow$ without $d \uparrow$
 $P \uparrow$ with $d \uparrow$

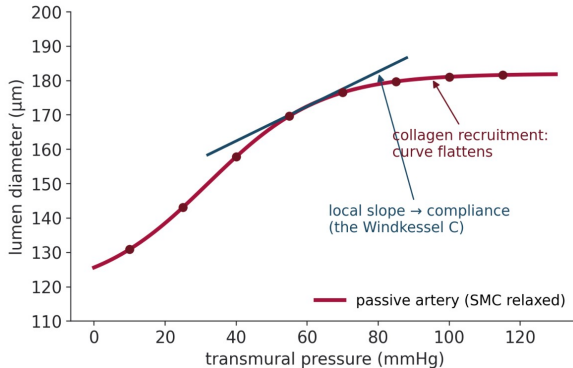
uncontrolled
stress held constant

Chronic pressure load $\rightarrow$ wall thickening $\rightarrow$ stress normalised. The tissue **defends its working stress**.

Occurs in the hypertensive aorta (medial thickening) and in the pressure-loaded left ventricle (concentric LVH).

The cost: more tissue to build and perfuse, and a stiffer vessel or chamber. Compensation is never free (HFpEF, Lecture 3).

Measuring wall mechanics: the pressure–diameter relation

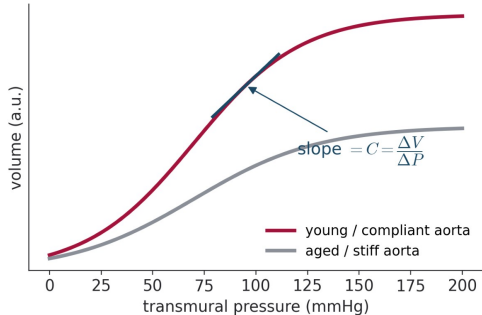


A small artery on two glass cannulas, smooth muscle relaxed (calcium-free bath), pressurised in steps while the diameter is tracked optically.

- compliant at low pressure — elastin bears the load
- flattening near physiological pressure — collagen recruited: the J-curve in P – D coordinates
- finite diameter at $P = 0$ — a moulded tube, not a balloon

The **local slope is the compliance**: a property of the working point, not a constant of the vessel.

Compliance: the functional summary



$$C = \frac{dV}{dP} \quad [\text{mL/mmHg}]$$

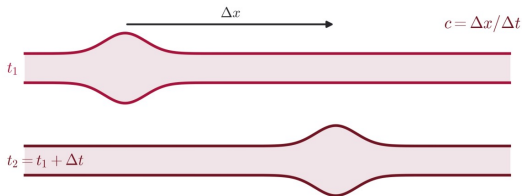
volume stored per unit transmural pressure: one parameter for all the wall mechanics so far.

- inherits the J-curve: falls as P rises
- concentrated in the proximal aorta
- **total arterial $C \approx 1\text{--}2 \text{ mL/mmHg}$**

Consistency check

$SV \approx 70 \text{ mL}$. If all of it were stored: $PP \approx SV/C \approx 40\text{--}70 \text{ mmHg}$ — the right order of magnitude. Storage is the principal event of systole.

The pulse propagates as a wave



Ejection distends the proximal aorta; the bulge stretches the next segment, which stretches the next.

Two speeds must be kept distinct:

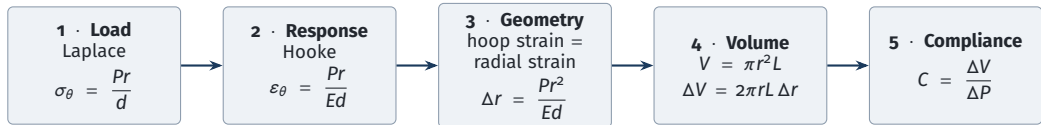
blood velocity	$\sim 0.2 \text{ m/s}$	the material
wave speed	$\sim 5 \text{ m/s}$	the disturbance

The pulse outruns the blood by a factor of ~ 25 : the radial pulse arrives at the wrist well before any molecule ejected in that beat.

Wave speed = **stiffness over inertia**: a stiffer tube transmits faster; denser blood slows it.

Anything that propagates can also **reflect**.

From wall composition to compliance, in five steps



$$C = \frac{2\pi r^3 L}{E d}$$

The exponents carry the physiology: $C \propto r^3$ — geometry dominates; $C \propto 1/E$, $C \propto 1/d$ — stiffening and thickening both reduce compliance; dilation raises it as the cube.

Check: $E = 0.5 \text{ MPa}$, $d = 2 \text{ mm}$, $r = 12 \text{ mm}$, $L = 0.5 \text{ m} \rightarrow C \approx 0.6\text{--}1 \text{ mL/mmHg}$ for the aorta alone.

Pulse wave velocity: two expressions, one law

Moens-Korteweg (1878)

$$c = \sqrt{\frac{Ed}{2\rho r}}$$

from the local wall composition

Bramwell-Hill (1922)

$$c = \sqrt{\frac{V}{\rho C}}$$

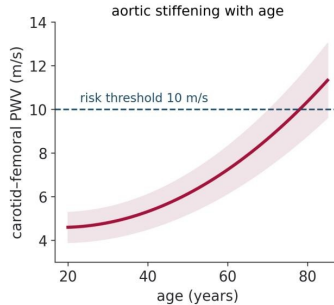
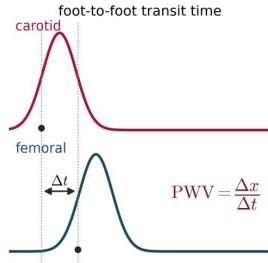
from measurable volume and compliance

Substitute $C = 2\pi r^3 L / (Ed)$ and $V = \pi r^2 L$ into Bramwell-Hill: $c = \sqrt{\frac{V}{\rho C}} = \sqrt{\frac{\pi r^2 L \cdot Ed}{\rho \cdot 2\pi r^3 L}} = \sqrt{\frac{Ed}{2\rho r}}$

Key idea

Forty-four years apart, one law: stiffness over inertia, as for every wave speed in physics. The speed of the pulse measures the modulus of the wall.

Carotid-femoral PWV: a stiffness meter in the clinic



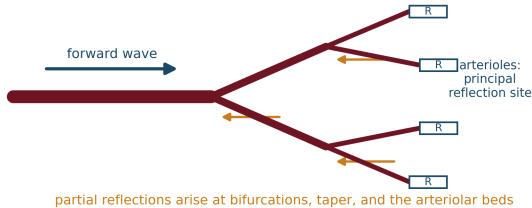
$$cfPWV = \frac{\text{path distance}}{\text{foot-to-foot transit time}}$$

The foot is used because the systolic peak is already contaminated by reflections.

young adult	~5 m/s
elderly	10–12 m/s
guideline threshold	> 10 m/s

Ageing raises E by $\sim 4\text{--}6\times$; since $c \propto \sqrt{E}$, the speed roughly **doubles**.

Wave reflection in the arterial tree



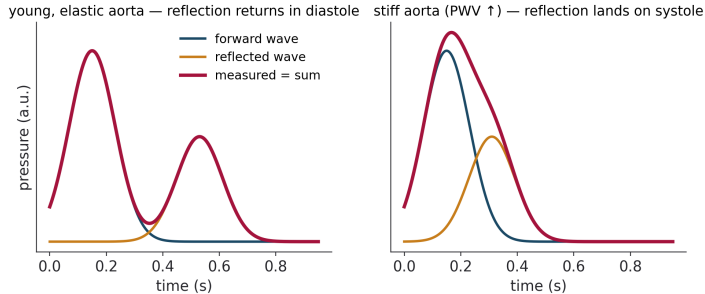
Any wave meeting a change of medium is partially reflected. Reflection sites: bifurcations, taper, changes of wall stiffness and — above all — the **high-resistance arterioles**.

No single reflected wave returns: a superposition of many partial reflections, behaving as though from an *effective* site $\sim 0.5\text{--}0.7$ m from the heart.

Key idea

Every measured waveform is the sum of a forward and a backward wave.

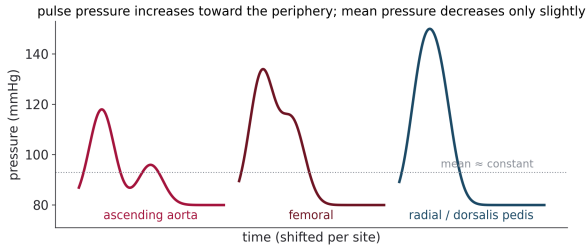
Forward and reflected waves: timing is everything



Young, elastic aorta ($c \approx 5$ m/s): round trip ~250–300 ms — the reflection arrives *after* the systolic peak and **raises diastolic pressure**. Beneficial: the coronaries are perfused in diastole.

Stiff aorta (c doubled): return in ~120–150 ms — the reflection lands *on* the systolic peak and **augments it**. Same heart, same stroke volume, higher systolic load — purely from timing.

Peripheral amplification and the ageing waveform



In the **young**, pulse pressure *increases* toward the periphery: the brachial PP exceeds the aortic — superposition geometry alone.

Ageing moves reflections into systole: central PP rises to meet the peripheral value.

The cuff measures the arm, not the aorta. The same brachial reading can mean a different central load — and central pressure is what the heart and brain experience.

The chain of Part 2, closed: composition → stiffness → compliance → wave speed → timing → the ageing waveform. Every step passed through the wall.

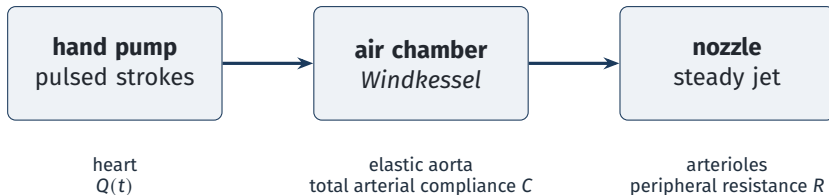
Checkpoint · Part 2

- 1 A patient stands up. Which of the three pressure differences change, and by how much at the ankle?
- 2 Derive $\sigma_\theta = Pr/d$ from a free-body diagram. Why does the projected area appear, and not the curved surface?
- 3 Why would a vessel with a linear stress–strain law fail at high pressure, and what feature of the real wall prevents this?
- 4 Using Moens–Korteweg, what happens to PWV if E quadruples? If the radius doubles at constant E and d ?
- 5 Why does the reflected wave assist coronary perfusion in the young but load the ventricle in the old?

Part 3 · The two-element Windkessel

Frank's analogy → storage and recoil → the electrical analogue → the governing ODE → validation → provable properties → Z_c and waves → limitations → FFR and allometry

The Windkessel analogy (Otto Frank, 1899)



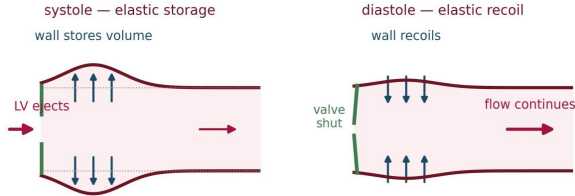
Frank's objective — still the model's use today: estimate cardiac output from pressure alone. Fit the model to a pressure trace, then infer the flow.

The essential step: some 10^{10} vessels compressed into two parameters — C from the wall (Part 2), R from the arterioles (Lecture 1).

Key idea

Expectation: very accurate in diastole and at low frequencies; blind to wave phenomena *by construction*. Both are demonstrated below.

Systolic storage, diastolic recoil



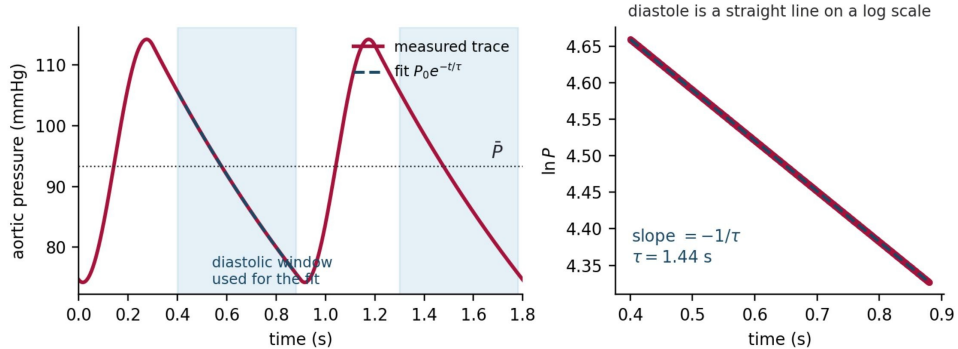
Systole. The heart ejects faster than the periphery drains; roughly half the stroke volume is stored in the distending aorta as elastic energy.

Diastole. Valve closed, inflow zero. The wall recoils and discharges the stored volume through R : flow to the tissues continues, pressure falls smoothly — exponentially, with $\tau = RC$.

Key idea

Without C , pressure would fall toward venous values between beats and capillary flow would be as pulsatile as ejection. The heart delivers pulses, the tissues require steadiness; the elastic aorta converts.

Worked example · fitting R and C to a pressure trace



Simulated aortic trace, 114/74 mmHg at 67 bpm ($T = 0.9 \text{ s}$), $SV = 70 \text{ mL} \rightarrow CO = 70 \times 60 / 0.9 = 4.67 \text{ L/min} = 77.8 \text{ mL/s}$. Only the pressure trace and CO are assumed measured; R and C are to be recovered.

Worked example · the three-step fit

Step 1 — the time constant

In diastole $Q = 0$, so the ODE reduces to $C \, dP/dt = -P/R$:

$$P(t) = P_0 e^{-t/\tau}, \quad \ln P = \ln P_0 - t/\tau$$

Straight-line fit to $\ln P$ over the diastolic window:

$$\tau = 1.44 \text{ s.}$$

Step 2 — the resistance

Over one cycle the capacitor returns what it took, so all mean flow crosses R :

$$R = \frac{\bar{P}}{CO} = \frac{93.3}{77.8} = 1.20 \text{ mmHg s/mL}$$

Step 3 — the compliance

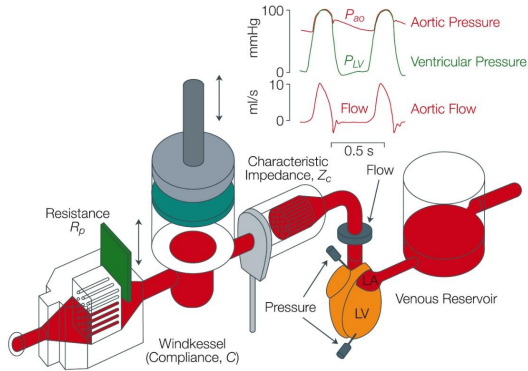
$$C = \frac{\tau}{R} = \frac{1.44}{1.20} = 1.20 \text{ mL/mmHg}$$

recovering the values used to generate the trace.

Consistency: $PP \approx SV/C = 70/1.2 \approx 58 \text{ mmHg}$ vs. 40 measured — an upper bound, since a third of SV drains through R during ejection.

Cautions on real data: exclude the first ~100 ms after valve closure (incisura, residual reflection); τ is sensitive to the window — the dominant uncertainty in C .

Experimental realisation: the isolated-heart preparation



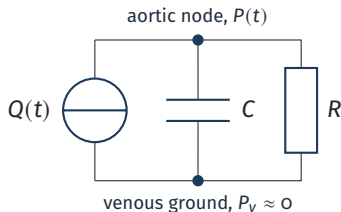
After Westerhof et al., *Snapshots of Hemodynamics*, 2nd ed., Fig. 24.1

An isolated ejecting heart pumps into a hydraulic model instead of a real aorta: lumped peripheral resistance, lumped compliance, and the characteristic impedance Z_c of the proximal aorta — in hardware.

The decisive test: the pressures and flows the heart generates against this artificial load are **indistinguishable** from those in the intact animal.

The same principle underlies the **mock circulatory loop**, the standard bench for testing valve prostheses, ventricular assist devices and catheters before any animal or clinical study.

The electrical analogue



Hydraulic	Electrical	Relation
pressure P	voltage U	across variables
flow Q	current I	through variables
resistance (arterioles)	resistor R	$P = QR$
compliance (aorta)	capacitor C	$Q_C = C \, dP/dt$
inertance (mass)	inductor L	$\Delta P = L \, dQ/dt$
volume conservation	Kirchhoff's law	node balance

Topology. C and R are in *parallel*: every millilitre ejected either distends the aorta (enters C) or crosses the periphery (through R) — one node, two exits. Stiff aorta = small capacitor; vasodilation = smaller R .

Deriving the governing equation · steps 1–3

Step 1 — the balance. In dt the heart delivers $Q(t) dt$ to the aortic node. Two destinations, nothing accumulates at the node itself:

$$Q(t) = Q_C + Q_R$$

Step 2 — the storage term. Compliance $C = dV/dP$, i.e. $dV = C dP$; divide by dt :

$$Q_C = \frac{dV}{dt} = C \frac{dP}{dt}$$

proportional to the *rate of change* of pressure: a wall that is not moving stores nothing, however distended.

Step 3 — the drain term. Ohm's law of the circulation, Poiseuille aggregated over the microcirculation, with $P_V \approx 0$:

$$Q_R = \frac{P - P_V}{R} \approx \frac{P}{R}$$

proportional to pressure itself, instantaneous: no memory.

Deriving the governing equation · assembly and diastole

Step 4 – assembly.

$$C \frac{dP}{dt} + \frac{P}{R} = Q(t)$$

A linear first-order ODE for $P(t)$: first-order — one energy store; linear — both constitutive laws linear; constant coefficients — R , C fixed over a beat. Every term is a flow (mL/s).

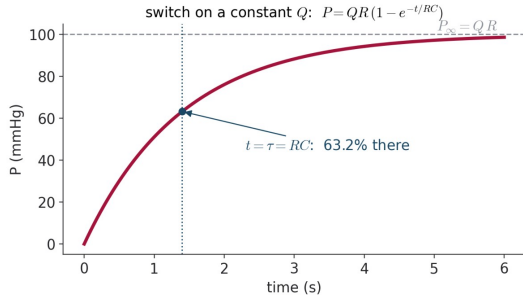
Step 5 — the diastolic solution. Valve shut, $Q = 0$:

$$C \frac{dP}{dt} = -\frac{P}{R} \implies \frac{dP}{P} = -\frac{dt}{RC} \implies P(t) = P_0 e^{-t/\tau}, \quad \tau = RC$$

Key idea

RC is the *only* time scale the model possesses. $R \approx 1.2$ mmHg·s/mL, $C \approx 1.2$ mL/mmHg give $\tau \approx 1.4$ s $>$ $T \approx 0.9$ s: the aorta never discharges fully within a beat — why diastolic pressure stays near 75 mmHg rather than falling to zero.

Validation I · the step response and its two limits



Constant flow Q_0 switched on at $t = 0$, $P(0) = 0$:

$$P(t) = Q_0 R (1 - e^{-t/\tau})$$

$t \rightarrow 0$: $e^{-t/\tau} \approx 1 - t/\tau$, so

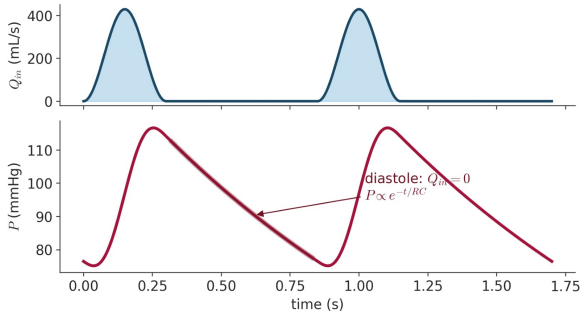
$$P \approx \frac{Q_0}{C} t, \quad \left. \frac{dP}{dt} \right|_0 = \frac{Q_0}{C}$$

the initial slope depends on **C alone** — nothing drains yet: early systole.

$t \rightarrow \infty$: $P(\infty) = Q_0 R$ — **R alone**: capacitor charged, Ohm's law recovered.

After one τ : 63.2% of the excursion; within 2% of plateau by 4τ . This curve is the first validation gate of the practical's forward model.

Validation II · pulsatile inflow



Input: a half-sine ejection in systole, zero in diastole. The same ODE produces a recognisable aortic trace:

- systolic rise while inflow exceeds the drain
- peak near end-ejection
- smooth exponential diastole

Two parameters reproduce a patient's waveform to within a few mmHg — the practical justification for lumped models.

Not reproduced: no incisura (no valve), no reflected-wave augmentation (no waves) — the systematic residuals of the practical's Part D.

Provable properties

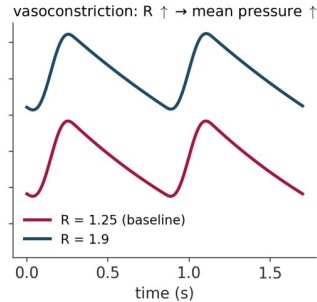
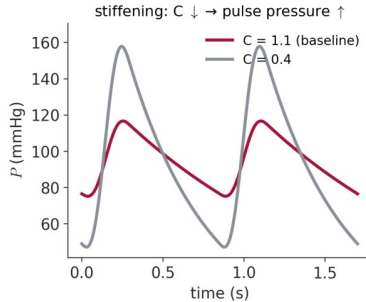
- 1 Low-pass filter.** Pulsatile flow in, smooth pressure out. Corner frequency $1/(2\pi\tau) \approx 0.1$ Hz, far below the heart rate (~ 1 Hz): the fundamental passes attenuated, the harmonics are suppressed.
- 2 Both elements are necessary.** C alone integrates the inflow indefinitely; R alone reproduces the input waveform exactly ($P = QR$). Smoothing needs the pair.
- 3 Over one period the capacitor nets to zero:** $\oint C dP = C[P(T) - P(0)] = 0$. All mean flow crosses R :

$$\bar{P} = CO \times R$$

Key idea

Mean pressure belongs to R ; pulse pressure belongs (mostly) to C : to first order $PP \approx SV/C$.
Two parameters, two distinct hypertension.

Two parameters, two forms of hypertension

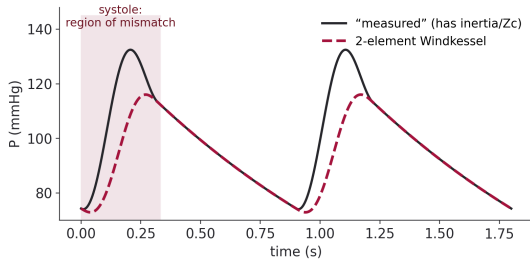


$C \downarrow$ **at constant R** — mean preserved, pulse pressure widened: **isolated systolic hypertension**, the ageing phenotype.

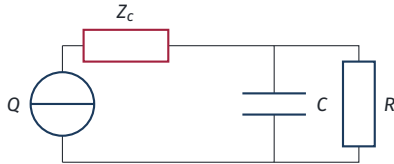
$R \uparrow$ **at constant C** — the whole curve shifts upward: **essential hypertension**.

Real patients move both at once — yet the two signatures stay separable: reading SBP/DBP/PP against these patterns is reading R and C from the cuff. The measured impedance spectra, three slides on, show both changes directly.

The three-element Windkessel: characteristic impedance



Where does the two-element model fail first? At the onset of flow: it lets pressure rise only as fast as C fills, whereas the real aorta responds instantly with a small proportional jump — the inertance of Part 1, expressed as an impedance.



Validity and limitations

Trust it for

- mean pressure
- the diastolic decay, $\tau = RC$
- estimation of total C and R
- the lowest harmonics ($\lesssim 2$ Hz)
- the afterload seen by heart models (Lecture 3)

Do not ask it for

- wave travel and reflection (PWV, augmentation)
- pressure differences along the tree — one pressure only
- systolic fine structure; anything above ~ 3 rd harmonic

Limitation

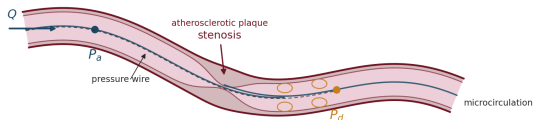
inertia ignored — systole described less well
proximal viscous loss ignored
aorta lumped to a point — it fills sequentially

$Q(t)$ imposed — blind to afterload

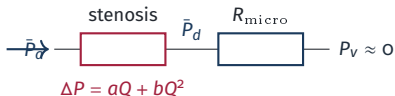
Correction on the course map

add L or Z_c (three-element model)
measured input impedance
wave propagation and reflection (Part 2); the practical's fit residuals
couple a heart model, $Q(P_{a.o})$ (Lecture 3)

Application · coronary stenosis and fractional flow reserve



$$\Delta P_{sten} = aQ + bQ^2 \quad \text{---} \quad FFR = \bar{P}_d / \bar{P}_a \text{ at maximal vasodilation}$$



The stenosis of Part 1 as a series element: a from viscous drag, b from exit separation.

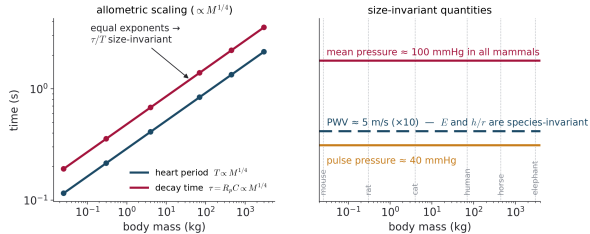
At rest Q is low, both terms small — asymptomatic. **Vasodilate** the microcirculation (adenosine: pharmacological exercise): Q rises, the bQ^2 term rises quadratically, $\bar{P}_d$ falls sharply.

$$FFR = \frac{\bar{P}_d}{\bar{P}_a} \text{ at maximal vasodilation;} \quad FFR \leq 0.80 \Rightarrow \text{revascularise}$$

A pressure wire and two averaged values decide whether an intervention is performed.

Coronary specificity: the myocardium compresses its own vessels in systole, so coronary flow is mainly diastolic — one more reason the Windkessel's maintained diastolic pressure matters.

Comparative physiology: invariants from mouse to elephant



After Westerhof et al., *Snapshots*, 2nd ed., ch. 30

Two time scales scale identically with body mass M :

$$T \propto M^{1/4}, \quad \tau = RC \propto M^{1/4}$$

so their ratio is size-invariant.

Since $\bar{P}/PP \approx \tau/T$ (Windkessel algebra, Exercise 8), every mammal shares essentially the same arterial pressure: $\sim 100/40$ mmHg for mouse and elephant alike.

PWV is likewise invariant (~ 5 m/s): evolution has conserved E and d/r . A mouse is a human at 600 bpm.

Key idea. When the dimensionless ratios of a model are conserved across five orders of magnitude in body mass, the model has captured something physically real.

Checkpoint · Part 3

- 1 Derive the Windkessel ODE from conservation of volume at the aortic node, stating the constitutive law of each element.
- 2 A patient's diastolic decay has $\tau = 2.1$ s; mean pressure and cardiac output give $R = 1.4$ mmHg·s/mL. What is their total arterial compliance?
- 3 Why does mean pressure depend only on R and not on C ? Give the one-line proof.
- 4 Which failures of the two-element model does Z_c correct, and which does it not?
- 5 Why must FFR be measured at maximal vasodilation and not at rest?

Summary

Three central results · numbers to carry · what comes next

Three central results

Navier-Stokes

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g}$$

Four equations, four fields, nonlinear through the convective term. The classical laws are its limiting cases. Two groups: $\alpha = r\sqrt{\omega\rho/\mu}$ (unsteady inertia), Re (convective inertia). Aorta $\alpha \approx 14$; microvessels $\alpha \ll 1$.

The wall chain

$$\sigma_\theta = \frac{Pr}{d}, \quad C = \frac{2\pi r^3 L}{Ed}, \quad c = \sqrt{\frac{Ed}{2\rho r}}$$

Laplace loads the wall; the J-curve is its stability condition; Laplace $\rightarrow$ Hooke $\rightarrow$ geometry gives C; compliance and inertia set the wave speed (~ 5 m/s young); reflection timing shapes and ages the waveform.

The Windkessel

$$C \frac{dP}{dt} + \frac{P}{R} = Q(t), \quad \tau = RC$$

Conservation of volume at one node. Diastole decays with τ ; mean pressure belongs to R, pulse pressure mostly to C. Blind to waves by construction — legitimate because $\lambda \approx 4.5$ m at the heart rate.

Numbers to carry

Constant / relation	Value
blood density ρ	1060 kg/m ³
blood viscosity μ	3.5×10^{-3} Pa·s
1 mmHg	133 Pa
α aorta / arteriole	≈ 14 / ≈ 0.02
Re aorta at rest	≈ 1400 (laminar; $Re_c \approx 2200$)
wall E	≈ 0.5 MPa (elastin) $\rightarrow$ $\sim$ GPa (collagen)
aortic σ_θ	≈ 80 kPa at $\sim 15\%$ strain
total arterial compliance	1–2 mL/mmHg
PWV young / old	~ 5 / 10–12 m/s (threshold > 10)
$\tau = RC$	≈ 1.4 s ($> T$: the aorta never empties)
Z_c	$\approx 7\%$ of R
oscillatory power fraction	$\approx 15\%$ systemic (25% pulmonary)
FFR threshold	≤ 0.80